

SHERRY HE

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EDUCATION

Brown University

M.S. Computer Science — GPA: 4.0/4.0

Tufts University

B.S. Computer Science and Mathematics — GPA: 3.7/4.0

Sep 2024 - May 2026

Providence, RI

Sep 2020 - May 2024

Boston, MA

TECHNICAL SKILLS

Programming Languages: Java, Python, C++, SQL, Swift

Tools: Docker, Request, Matplotlib, OpenSearch, Pandas, BeautifulSoup, Selenium, AWS, IntelliJ Idea, WebStorm

Front-end: React, Vue, Node.js, HTML, CSS, JavaScript

Back-end: Springboot, SSM, MySQL, Redis, Kafka, Elasticsearch

Machine Learning: CNN, ResNet, Model Evaluation

EXPERIENCE

Haier Smart Home

June 2025 – August 2025

Software Engineer Intern

- Designed and implemented **RESTful APIs** for IoT device control using **Spring Boot**, supporting **10K+** concurrent requests with **99.9%** uptime. Integrated **Zigbee/Z-Wave** communication protocols for cross-platform device.
- Developed automated testing framework (**Pytest + Selenium**) reducing regression test time by **40%**.
- Migrated legacy monolithic architecture to **Kubernetes** cluster, improving horizontal scaling capabilities. Optimized **PostgreSQL** query latency from **1.2s** to **200ms** via indexing strategies and connection pooling
- Automated deployment pipeline using **Jenkins/GitHub Actions**, cutting release cycles from **6h** to **45min**.
- Implemented **Docker** containerization for microservices, improving environment consistency across dev/staging.

Zaiqi Tech

June 2024 – August 2024

Software Engineer Intern

- Designed lightweight edge computing nodes for IoT sensor data preprocessing, reducing cloud latency by **40%** through protocol optimization (**MQTT** over **QUIC**)
- Architected **Flink**-based analytics engine processing **200K+** events/sec with stateful computations.
- Migrated legacy monolithic API gateway to **gRPC** microservices and implemented service mesh (**Istio**) for traffic management, reducing API error rates by **68%**.

PROJECTS

Social Media app | MongoDB, Express, React, Node.js, Redux

Sep 2024 - Dec 2024

- Developed the app with **MERN (MongoDB, Express.js, React.js, Node.js)** for flexibility and scalability. It supports features such as **Google login, posting, commenting, searching, and recommendations**.
- Implemented **Test-Driven Development (TDD)** to increase code test coverage and enhance code quality. Introduced **GitLab** to design robust **CI/CD** pipelines to automate the code testing and integration process.
- Enhanced **MongoDB** query performance for searching posts by strategically creating **indexes** and implementing **pagination**, resulting in a **31%** reduction in response time.
- Implemented the **express-rate-limit rate limiting component** to protect the system and improve system stability.
- Integrated **DFA (Deterministic Finite Automaton) algorithm** for sensitive word filtering, reducing the content risk rate of user posts by **75%**.
- Successfully deployed the server on **AWS EC2** (Amazon Web Services) to handle more than 125 queries per second.

Agentic Earth Monitoring System | Javascript, FastAPI, Kafka, CrewAI, AWS

October 2025

- Built event-driven architecture using **Node.js/Express** and **Kafka** for ingesting **10K+** satellite image streams/minute.
- Developed data transformation workflows with **AWS Lambda** and **FastAPI** for geospatial data normalization.
- Integrated **CrewAI** framework for multi-modal data processing (satellite imagery + IoT sensor data).
- Designed model orchestration layer supporting **ONNX Runtime** and **TensorRT** inference acceleration.
- Architected **Kubernetes (EKS)** cluster for container orchestration and auto-scaling.
- Built **CI/CD** pipeline with GitHub Actions and Docker for automated deployments.

InvMicroServices | Java, REST, Docker, Kubernetes

Apr 2023 – Jun 2023

- Designed and developed a MicroService system based on **REST API** functionality, serving as an autograder to manage code submissions and perform **evaluations of code execution** for more than a **thousand** of users.
- Implemented an end-to-end **CI/CD** pipeline using **Jenkins** for building and testing, leveraging **Docker** and **Kubernetes** for efficient container orchestration and automated deployment. This strategy significantly reduced maintenance overhead and accelerated development iterations.
- Optimized database interactions using **Hibernate's** built-in cache, enhancing data handling performance and operational efficiency by **20%**.
- Leveraged **RabbitAdmin, RabbitTemplate, and RabbitListener** to establish an efficient asynchronous communication mechanism between MicroServices, leading to a **30%** increase in system throughput.